

# Submission — Decoding Customer Value: A SQL-Driven Retention Strategy

Summer Projects '26 · Consulting & Analytics Club, IIT Guwahati · Submitted by Raman Mishra

## The answer (read this first)

Is the brand building a loyal base, or reliant on promotions? **Neither story is true.** The brand has a genuinely committed base it is failing to *activate*; promo reliance is modest and concentrated, while under-activation is large and fixable.

- **15.8% of revenue (\$36,766)** is plausibly discount-caused — the entire measurable promo exposure. Discount dependency is **not** the core problem.
- **26.9% of revenue (\$62,645)** flows through subscriber discounts whose necessity is untested by design.
- **969 "committed-but-slow" customers (25% of base)** carry a **~\$996k value gap** to their activated equivalents — **2.7x the size** of the measurable promo exposure, and the single largest opportunity.

Loyalty was **defined, not declared** — two competing definitions (behavioral vs. commitment-based) built, tested, and the commitment definition adopted; an automated definition-search with a planted-signal control confirmed no data-mined alternative beats it.

## Deliverables in this folder

| #  | File                                                                             | What it is                                                                                                                                            | Satisfies            |
|----|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| D1 | D1 - Engineered Data.xlsx · D1 - Feature Engineering (Python)/                   | Cleaned dataset + engineered features (dependency score, value tier, satisfaction flag); the 6 Python scripts that build them, with logic per feature | Python               |
| D2 | D2 - Segmentation Queries.sql · D2 - SQL Results/                                | Segmentation queries answering all 5 key business questions, plus the exported result table for each                                                  | SQL                  |
| D3 | D3 - Founder Dashboard (Power BI)/ + D3 - Founder Dashboard (static preview).pdf | Four-panel founder dashboard (pyramid · segments · geo · category) + a static PDF preview that opens without Power BI                                 | Power BI             |
| D4 | D4 - Retention Playbook.pdf                                                      | Promotional sunset plan + ideal customer profile, each with named segment, trigger, timeline, tracked metric, and trade-off                           | Playbook             |
| D5 | D5 - Executive Summary.pdf                                                       | One-page findings + recommendations under either scenario                                                                                             | Summary (max 1 page) |

## How to open

- **D1** — **D1 - Engineered Data.xlsx** opens in Excel; the Python scripts run in order (**01\_clean** → **06\_playbook\_numbers**).
- **D2** — open the **.sql** in any client, or read pre-exported answers in **D2 - SQL Results/**.
- **D3** — **D3 - Founder Dashboard (static preview).pdf** opens anywhere (no install); for the interactive version open **RetentionDashboard.pbip** in Power BI Desktop (free).
- **D4 / D5** — PDFs, zero-install.